

CONTACT



+90 531 971 03 39
+421 901 79 00 34



utkukaraca.sql@gmail.com



Bratislava, Slovakia



[Portfolio Website](#)

EDUCATION

TOBB UNIVERSITY OF ECONOMICS AND TECHNOLOGY

2017–2022

B.Sc. in Electrical and Electronics Engineering
Ankara, Turkey

Ege University – International Computer Institute

M.Sc. in Computer Engineering | 2024–2026

Research focus: Algorithm design and modeling in imperfect network environments; distributed systems coordination
İzmir, Turkey

LANGUAGE

- English (Professional working proficiency)
- German (Basic)

TECHNICAL SKILLS

Programming

- C, C++, Python, Java, TypeScript/JavaScript, Bash, SQL, Kotlin

Backend & Integration

- REST API Development, Angular, Apache Tomcat, JDBC, Data Pipelines.

Databases

- Oracle, PostgreSQL, MS SQL Server, SQLite, Database Migration, Query Optimization, Schema Design.

Software Engineering

- Distributed Architectures, Concurrency & Multithreading, Inter-Process Communication (IPC), Algorithm Design, Reverse-Engineering.

PROFILE

Software developer specializing in backend systems, distributed architectures, and large-scale software platforms. Currently working across both Project and R&D domains for a global leader in critical infrastructure technology. As the creator and core developer of key integration modules, I focus on developing the interfaces that enable mission-critical operational platform connectivity with external resources, while contributing to base-level development for distributed systems across various projects.

Experienced in designing plugin-based architectures and integration tooling that facilitate reliable communication between modern services and production environments, with a strong foundation in applying predictive modeling and AI-assisted data estimation tools. Strong background in debugging large-scale systems and optimizing performance to ensure high reliability across distributed software platforms.

Motivated by solving complex engineering challenges and currently pursuing an M.Sc. in Computer Engineering with research focused on distributed algorithms in imperfect networks.

WORK EXPERIENCE

SIEMENS

February 2023 – Present

SOFTWARE DEVELOPER

GRID SOFTWARE/ SCADA Software

R&D: Enterprise-Scale Core Platform Extensions & Cloud Integration

- Acted as the creator and lead developer of interface bridges enabling bidirectional data transfer between the flagship core operational platform and external cloud SaaS applications.
- Designed and implemented mission-critical integration bridges for domain-specific core services, including secure switching, telemetry calculations, and real-time alarm interfaces.
- Engineered REST API-based communication protocols to modernize legacy infrastructure, transforming the system into a modular, plugin-ready ecosystem for bidirectional data exchange with external sources (Kafka etc.).
- Built and programmed custom R&D software components from scratch, ensuring high-performance data transfer and reliability within complex operational control environments.
- Integrated robust validation mechanisms across the new interfaces to ensure data integrity and full compatibility with existing core system data models.
- Executed system-level development to deliver maintainable and extensible solutions within complex distributed software architectures.

SCADA Reporting Tool

- Developed an end-to-end full-stack reporting tool from scratch as the primary creator, enabling clients to efficiently extract and generate customized operational reports.
- Engineered a full-stack application utilizing an Angular-based user interface and a Java backend, deployed on Apache Tomcat servers.
- Established robust JDBC connections to interface directly with time-series historical databases, facilitating the reliable retrieval and processing of large-scale client datasets.
- Integrated advanced Machine Learning and Predictive Modeling features to perform real-time estimation and automated recovery of missing telemetry data, ensuring high accuracy and continuity in operational reports.
- Strengthened system cybersecurity by integrating Active Directory (AD) for robust authentication and access control, ensuring secure communication and strict data protection.
- Streamlined the client experience by delivering a flexible, customizable platform that simplifies complex data extraction and customized report generation.

Tools & Environment

- Git, Docker, Linux/Unix, Maven, Gradle, GDB, DBX.

SCADA & Systems

- Operational control systems, telemetry data handling, historical data processing, distributed systems, plugin-based extensions

AI & Machine Learning

- LLM & Generative AI, Prompt Engineering, Agent-Based AI Systems, Deep Learning (CNN, RNN, Transformers), Neural Network Design, Predictive Data Modeling, Time-Series Estimation, Automated Data Imputation

Cybersecurity & Identity

- Active Directory (Authentication & Access Control), Applied System Security, Vulnerability & Security Fundamentals

Internships

Siemens – Smart Grid & SCADA Systems
Energy Systems Software Team
Sep 2021 – Dec 2021

- Supported development and testing activities related to smart grid monitoring and SCADA software components
- Assisted in system analysis, debugging, and documentation of operational workflows

TOBB Economics and Technology

University - Research Project
Jan 2021 – Apr 2021

- Developed machine learning models for data analysis and pattern recognition tasks
- Implemented deep learning approaches and evaluated model performance on experimental datasets

Portfolio

[LeetCode](#)
[Github](#)
[Tryhackme](#)
[Hackerrank](#)

Advanced Operational Reporting & Monitoring Platform

- Acted as the sole creator and developer of a custom real-time reporting and visualization platform for large-scale distributed systems, building the application entirely from scratch.
- Engineered a complex data architecture that captures real-time relational telemetry data and communicates directly with the core communication server, operating natively within the system kernel.
- Integrated dual-database querying capabilities, successfully extracting, processing, and correlating extensive datasets from both time-series historical databases and fault management systems.
- Developed an advanced User Interface (UI) featuring interactive network mapping and a suite of ready-made, customizable templates tailored to meet complex client reporting requirements.
- Implemented comprehensive live monitoring to display real-time alarms and system messages, introducing the capability to generate sophisticated, indirect reports derived directly from these operational events.

Core Platform Engineering & Specialized Integrations

- External Telemetry & Redundancy Systems: Acted as the primary creator of custom integration solutions using Python-based ingestion pipelines and proprietary scripting to inject third-party telemetry data; simulated native hardware behavior and engineered automated redundancy mechanisms to ensure zero data loss and continuous data flow.
- Legacy Data Migration & Modernization: Engineered and executed a fully automated pipeline to migrate 20 years of mission-critical historical data from obsolete systems to modern time-series database architectures; managed the end-to-end process including complex data transformation, schema mapping, and rigorous validation within a two-month timeframe.
- System-Level Enhancements & Debugging: Owned the development of project-specific core features and multi-site configurations within the flagship platform; conducted in-depth root-cause analysis and system-level debugging across distributed components to ensure seamless integration of new modules into high-reliability production environments

PROJECTS & RESEARCH

etdmnet – Open-Source P2P Networking Library (Kotlin/JVM & Android)

github.com/SqLkk/etdmnet | Apache 2.0 | 2025–2026

- Architected and published a data-relay-free peer-to-peer library enabling direct device communication over WebRTC DataChannels, eliminating the need for relay servers.
- Engineered the ETDM automatic host election and migration protocol based on EWMA link-quality scoring, directly implementing mathematical models from my M.Sc. thesis research.
- Designed a complex six-module Gradle architecture with full cross-platform support (JVM, Android, iOS preview) and established a complete CI/CD pipeline via GitHub Actions and JitPack.

PDA – Multi-Agent Orchestration Framework (In Development) Personal Research Project | 2025–Present

- Designing an LLM-powered multi-agent orchestration system featuring structured tool use, active memory management, and autonomous inter-agent communication protocols.
- Exploring and integrating advanced agent planning strategies (ReAct, chain-of-thought prompting) alongside custom neural pipeline components for complex task decomposition.